

SZABIST
UNIVERSITY
Islamabad Campus

Final Year Project Short Proposal

RetainIQ: Intelligent Customer Retention & Analytics System

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Introduction

RetainIQ is a web-based intelligent customer retention and analytics system designed to help businesses understand customer purchasing behavior, identify customers who are at risk of churn, and support data-driven retention decisions. Many businesses collect customer and transaction data but lack a unified system that converts this data into actionable retention insights. RetainIQ addresses this gap by combining customer analytics, RFM analysis, customer segmentation, machine learning-based churn prediction, risk scoring, explainable analysis, and retention recommendations in one platform.

Purpose

The purpose of RetainIQ is to provide a practical analytics platform that transforms raw customer transaction data into meaningful customer profiles and retention insights. The system will preprocess and validate uploaded data, calculate customer purchase metrics and RFM scores, segment customers, predict churn risk using machine learning, explain important risk factors, and allow users to explore possible changes through a What-If Risk Simulator. It will also provide retention recommendations and an interactive analytics dashboard.

Targeted Audience

The system is intended for small and medium-sized businesses, e-commerce operators, marketing teams, sales teams, customer-success staff, and business analysts who need to monitor customer behavior and prioritize retention efforts. It is designed for users who may have transactional data available but need an accessible analytical interface rather than a highly specialized data-science environment.

Product functions and features

RetainIQ will include customer and transaction data management, preprocessing and validation, purchase analytics, RFM analysis, customer segmentation, churn prediction, retention risk scoring, explainable customer risk analysis, Customer 360° profiles, a What-If Risk Simulator, retention recommendations and intervention planning, and a retention analytics dashboard and reporting module. The system will use machine learning to estimate churn probability while presenting interpretable risk factors so that predictions can support practical business decisions rather than functioning as unexplained scores.

Social and Industrial impact

RetainIQ can help businesses identify declining customer engagement earlier and allocate retention efforts toward customers who need attention. By combining predictive analytics with interpretable customer-level insights, the system can support more efficient marketing and customer-success decisions, reduce dependence on manual spreadsheet analysis, and encourage evidence-based retention strategies. Academically, the project demonstrates the integration of data preprocessing, machine learning, analytics, explainability, and web-based visualization into a single software system.



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Final Year Project (BS Artificial Intelligence)

Supervisor Consent Form

I accept the supervision of the following students with details as follows

Project Title:	RetainIQ:Intelligent Customer Retention & Analytics System
Student#1 Name & Registration Number	Muhammad Fahad Akram (23108173)
Student#2 Name & Registration Number	Muhammad Shoaib Ali (23108175)
Student#3 Name & Registration Number	Muhammad Mahad Atique (23108257)

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Signature: _____

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